

Green subsidies with demand distortions

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Climate change mitigation technologies

- ▶ Initial investment plus follow-through decisions
 - ▶ Buy an EV plus drive the EV
- ▶ Private and social benefits often positively correlated
 - ▶ Drive a lot of miles: save more fuel, abate more CO₂
- ▶ Advantageous selection into adoption
 - ▶ Those who abate the most are willing to pay the most, adopt first

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- ▶ Motivates marginal taxes/subsidies (Pigou 1920)
 - ▶ Corrects investment and follow-through decisions

Principals use both MC and FC green technology subsidies

Marginal cost subsidy

Fixed cost subsidy

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Fixed cost subsidy

EV charging, electric cooking:




Electric Vehicle (EV) rate plans

Making sense of EV rate plans

Residential EV rates

You can lower your energy costs by enrolling in one of our EV rate plans:

- Home Charging EV2-A
- Electric Vehicle Rate Plan EV-B

Volvo Offers Swedish Buyers A Year Of Free Charging To Spark EV Sales

♦♦

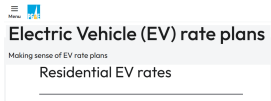
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BY NATALIE NEFF
 PUBLISHED: OCT 22, 2025 9:00 AM EDT

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Fixed cost subsidy

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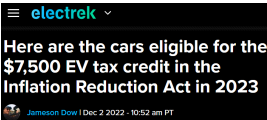
Heat pump grant increased to £7,500 by government in efforts to lower pollution and bills

Heat pumps emit far less pollution than gas boilers and can lower bills, but are expensive upfront. The government hopes the scheme will bring the cost of installing a new heat pump in line with that of a gas boiler.



Victoria Seabrook
Climate reporter @SeabrookClimate

Monday 23 October 2023 11:33, UK



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- ▶ We study demand distortions where $wtp_{\text{observed}} = (1 - \gamma) \cdot wtp_{\text{efficient}}$
 1. *Allocation distortion*: affects decisions (not payoffs)
 e.g. Inattention to benefits Dellavigna (2009), Present bias Laibson (1997); O'Donoghue, Rabin (1999); Indarte, Kluender, Malmendier, Stepner (2026)
 2. *Resource distortion*: affects decisions and payoffs
 e.g. High cost of administering credit

WTP distortions are pervasive in development

Our setting: credit market failures

- ▶ Solutions? Attempts: Microfinance, group lending, mobile transactions, digital collateral, etc.

Moral hazard and adverse selection persistently undermine credit markets:

- ▶ Informality: 1.3 billion adults lack a bank or mobile account World Bank (2025)
- ▶ Imperfect property rights: collateral
- ▶ Institutional constraints: savings (FDIC), nationwide credit bureaus (FICO), asymmetric borrower information
- ▶ Frictions in legal systems: debt recovery
- ▶ Relational economies: non-credible threats

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Banks respond by...

- Increasing the pecuniary and non-pecuniary costs of credit:
Higher interest rates, shorter grace and repayment periods, other-asset collateral, strict eligibility requirements, social penalties

Quinn & Woodruff (2019); Banerjee & Duflo (2005, 2014); Field et al. (2013); Kaboski & Townsend (2012); Jack et al. (2023); Fafchamps et al. (2014); Itskhoki & Moll (2019)

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We show that distortions can, in theory:

- ▶ Increase the welfare gain from fixed and marginal cost subsidies
- ▶ Increase the efficacy of fixed cost subsidies relative to marginal cost subsidies

Energy-intensive durable: Stoves

- ▶ Billions of people cook with biomass every day (usually wood or charcoal)
- ▶ This is usually a household's most polluting device: one charcoal stove emits approx. **4.9 tCO₂e** per year
- ▶ For comparison: driving a gasoline vehicle 11,500 miles at 22.2 MPG emits **4.6 tCO₂** per year EPA (2025)



This paper (2) Impact of induction stove subsidies in Kenya



Kenyan Jiko



Induction stove

Green technology subsidies: East Africa

Marginal cost subsidy



UMEME Customers Enjoy Cheaper Cooking Tariff Set By ERA

Christopher Kiiza · November 14, 2022 · 3 minutes read

eCooking Tariff Development

- Test the viability and impact of a specialized eCooking tariff for Tier 3+ connected households in Kenya

Cost: USD 450,000
(KES 57 million)

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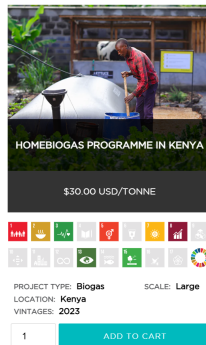
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Tax waivers on eCooking

- A dedicated study to define a suitable tax waiver regime to reduce upfront appliances costs to locally manufactured appliances, components and/or accessories, which will increase demand.
- Determine the economic and financial impact of these waivers on government revenues and household affordability.

Cost: USD 125,000
(KES 16 Million)



Randomized Controlled Trial

1. **Sample:** 2,134 Kenyan households
2. **Randomized treatments:**
 - ▶ **Marginal cost subsidies:** 0, 25%, or 75% off the cost of electricity used to cook with the induction stove
 - ▶ **Fixed cost subsidies:** 10% or 75% off the marginal cost of \$82
 - ▶ **Payment structure:** pay up front or pay with loan
3. **Measurement:**
 - ▶ 2min-frequency temperature monitors on charcoal stoves for 3+ months
 - ▶ 15min-frequency induction stove usage for 3+ months
 - ▶ In-person 1-month follow up survey

Preview: Impacts

1. The induction stove reduces monthly energy spending by \$8.9
 - ▶ \$209 over 2.2-year lifespan; 118% ROI on \$82 price
 - ▶ Monthly charcoal spending down \$10 (66%)
 - ▶ Monthly electricity spending up \$2.3 (65%)
2. Net abatement of 2.5 tCO₂e per year
 - ▶ For comparison: an EV abates between 0.1–1.5 tCO₂e per year

Muehlegger and Rapson (2023)

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3. Significant demand distortions:
 - ▶ Without a loan, average willingness-to-pay is \$20: $\bar{\gamma} = 0.91$
 - ▶ With a loan, average willingness-to-pay is \$35: $\bar{\gamma} = 0.83$

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4. Marginal cost subsidies have no impacts on adoption or usage
 - ▶ \$1 of subsidy raises WTP by $\leq \$0.2$
 - ▶ A 1% price decrease raises usage by $\leq 0.12\%$

Preview: How demand distortions affect subsidy efficacy

	Less distorted (loan)	More distorted (cash)
Observed willingness-to-pay	\$35	\$20

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	Less distorted (loan)	More distorted (cash)	Advantage of subsidies in a distorted market
Observed willingness-to-pay	\$35	\$20	
<i>Channel 1: Marginal adopters</i>			
Fuel savings per marginal adoption	\$175	\$228	+30%
Abatement per marginal adoption	4.8 tCO ₂ e	5.7 tCO ₂ e	+19%
<i>Channel 2: Subsidy cost</i>			
Subsidy spending per marginal adoption	\$93	\$65	−30%

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<i>Aggregate difference</i>			
Subsidy spending per tCO ₂ e abated	\$19	\$11	−41%

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Subsidy spending per tCO ₂ e abated	\$19	\$11	-41%
Observable welfare gain per \$1 spending			
Allocation distortion	\$7.19	\$12.76	+77%
Resource distortion	\$7.19	\$10.27	+43%

Preview: Model estimation and counterfactuals

- ▶ Our context still has significant demand distortions:
even with a loan, we estimate a distortion of $\bar{\gamma} = 0.83$
- ▶ What if demand distortions were reduced further?

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- ▶ What if demand distortions were reduced further?
- ▶ We use 2-step Generalized Method of Moments (GMM) to estimate 5 model parameters by matching to 7 moments observed in the experiment
- ▶ At $\bar{\gamma} = 0.91$, \$11 per ton of CO₂ abated
- ▶ Counterfactual subsidy spending under $wtp_{observed} = (1 - \bar{\gamma}) \cdot savings$:
 - ▶ At $\bar{\gamma} = 0.5$, \$73 per ton of CO₂ abated
 - ▶ At $\bar{\gamma} = 0$, \$162 per ton of CO₂ abated

Efficient carbon abatement at scale

- ▶ Annual carbon mitigation financing exceeds \$1 trillion Buchner, 2024
- ▶ Concerns about adverse selection and moral hazard
Einav, Finkelstein, and Cullen (2010); Jack, Jayachandran, Kala, Pande (2025); Aspelund and Russo (2026); Chen, Ryan, and Xu (2025); Azevedo, Berkouwer, Mishra, Russo, van Benthem (2026)
- ▶ Efficiency: Allocate marginal dollars to lowest cost abatement technologies

Green subsidies might abate more in distorted markets

- ▶ Weak institutions in many low- and middle-income countries
⇒ global capital misallocation
Acemoglu, Johnson, and Robinson (2005), Nunn (2007); Shapiro (2025)
- ▶ U.S.: 80% of new cars sold with financing ⇒ >\$1,200 per tCO₂e
Experian, 2023; Hahn et al., 2026; Allcott et al., 2026
- ▶ Kenya: APR 224% (even with digital collateral) ⇒ \$11 per tCO₂e

Interest rates

Today

Introduction

Theory

RCT Design

Impact

Externality of the marginal adopter

Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

Pared down version of model to build mechanisms intuition

- ▶ Each individual has inelastic energy services demand θ_i (e.g. cooking, driving)
- ▶ They can satisfy this demand with a dirty or greener version of a technology
- ▶ Benefits of clean energy:

$$\underbrace{\pi_i \equiv \theta_i(f_H - f_L)}_{\text{private (fuel savings)}}$$

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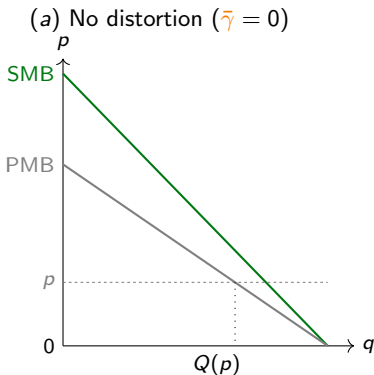
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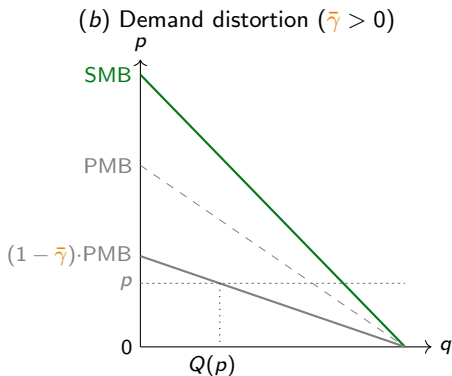
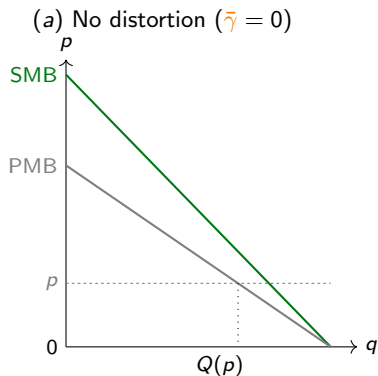
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1. $wtp_i = (1 - \bar{\gamma})\pi_i$: We focus on $\bar{\gamma} \in (0, 1)$
2. Market demand: $Q(p - s_1) = \sum_i \mathbb{1}[wtp_i \geq p - s_1]$

Distortions can increase abatement per fixed cost subsidy

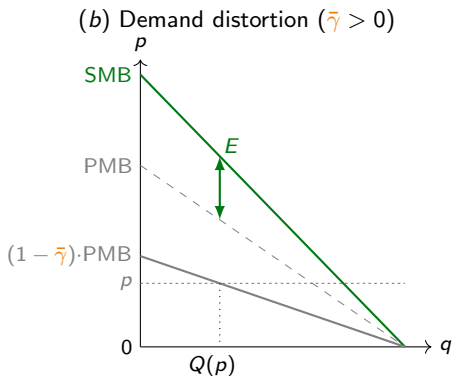
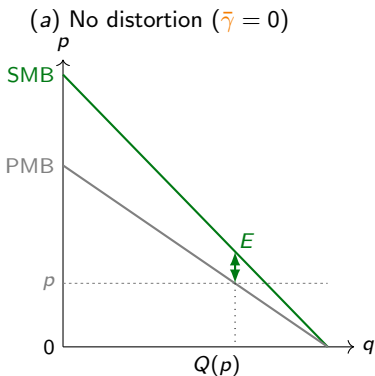


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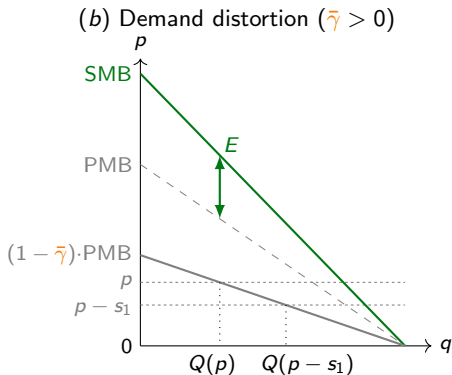
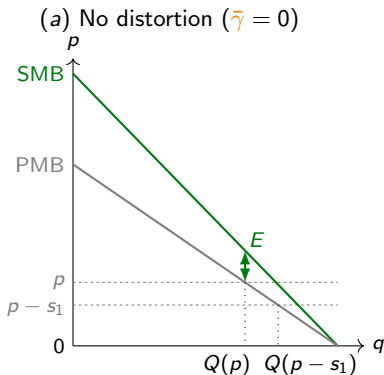
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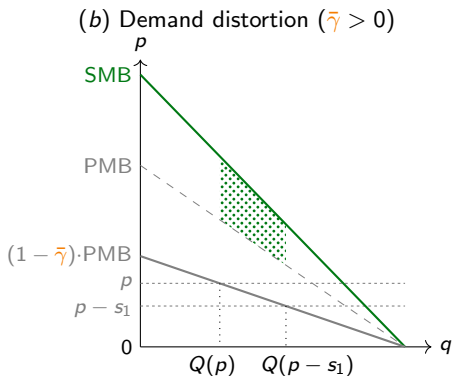
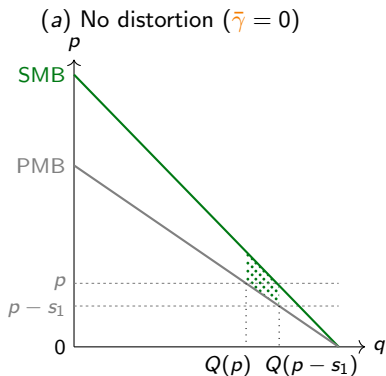
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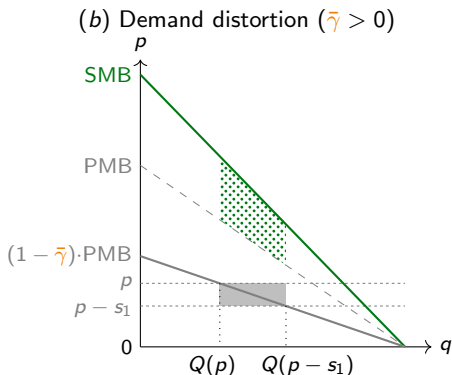
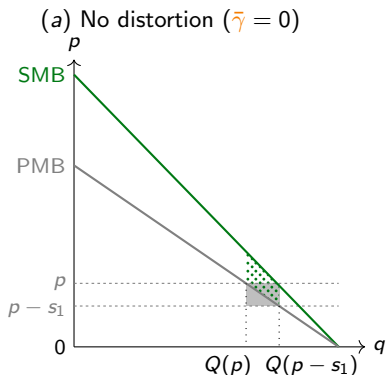
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 Environmental benefit

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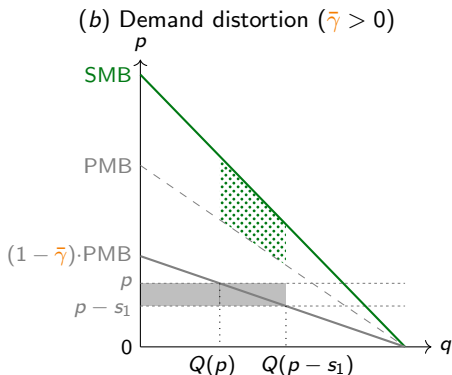
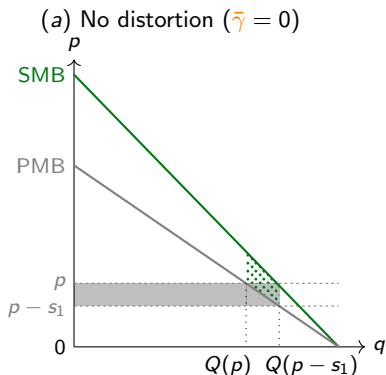
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 Environmental benefit
 Subsidy spending

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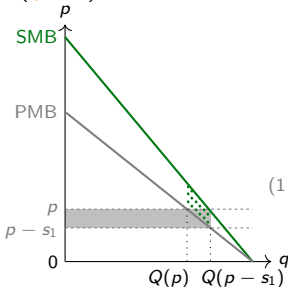
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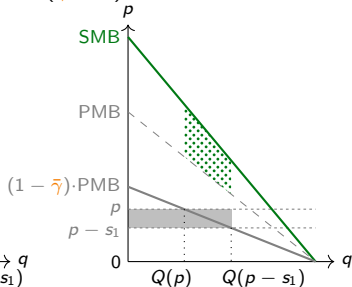
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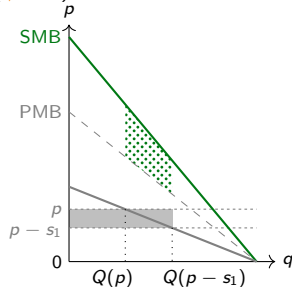
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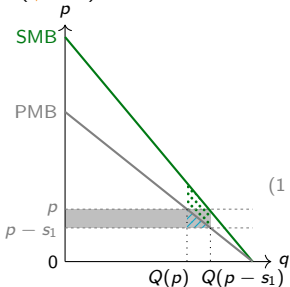


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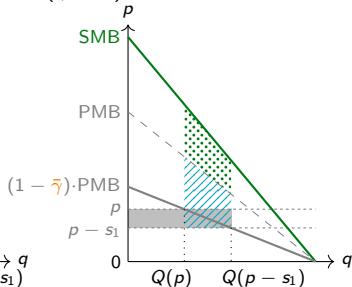
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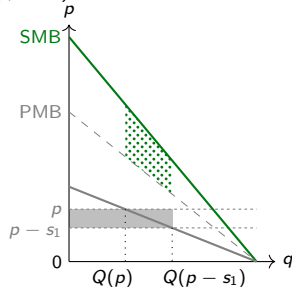
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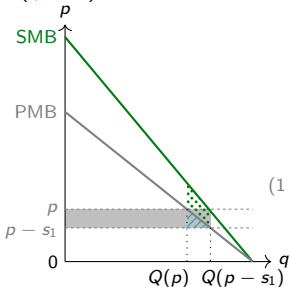
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Consumer surplus

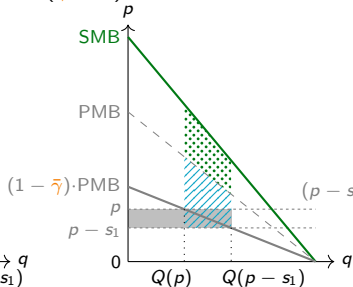
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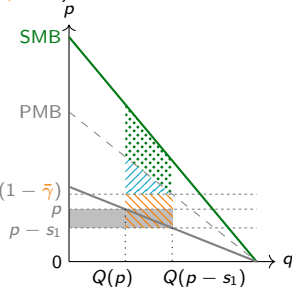
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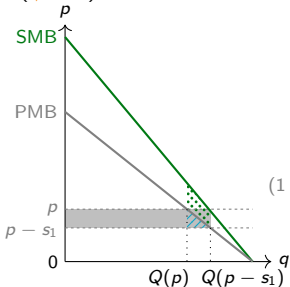
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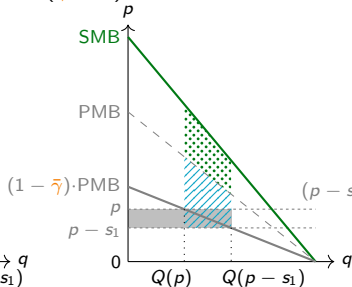
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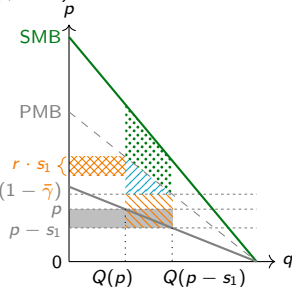
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Environmental benefit

Subsidy spending

Consumer surplus

Resource cost

Resource cost saved on inframarginal loans

Full model: Adoption and usage Full model

- ▶ Individual preference parameter α (e.g. learning, taste)
- ▶ Principal sets three policies:
 1. **Fixed cost subsidy** s_1 : purchase price discount
 2. **Marginal cost subsidy** s_2 : cleaner fuel price reduction
 3. **Marginal cost tax** τ : dirty fuel price increase
- ▶ Agents make two decisions:
 1. **Adopt**: buy the green technology at subsidized price
 2. **Usage**: fraction of energy services via green tech, generates usage cost

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 1. **Adopt**: buy the green technology at subsidized price
 2. **Usage**: fraction of energy services via green tech, generates usage cost
- ▶ Uniform demand distortions along two dimensions:
 1. Demand scaled down by $(1 - \bar{\gamma})$ - affects demand but not usage
 2. Perceived fuel savings dampened by δ - also affects usage [not today]

Full model: Adoption and usage Full model

- ▶ Individual preference parameter α (e.g. learning, taste)
- ▶ Principal sets three policies:
 1. **Fixed cost subsidy** s_1 : purchase price discount
 2. **Marginal cost subsidy** s_2 : cleaner fuel price reduction
 3. **Marginal cost tax** τ : dirty fuel price increase
- ▶ Agents make two decisions:
 1. **Adopt**: buy the green technology at subsidized price
 2. **Usage**: fraction of energy services via green tech, generates usage cost
- ▶ Uniform demand distortions along two dimensions:
 1. Demand scaled down by $(1 - \bar{\gamma})$ - affects demand but not usage
 2. Perceived fuel savings dampened by δ - also affects usage [not today]
- ▶ Throughout we assume standard bounds on derivatives and elasticities
Allcott, Mullainathan, Taubinsky (2014)

Optimal policy: Constrained principal

- ▶ **No authority to tax:** can subsidize purchase price (s_1) or usage price (s_2)
 - ▶ Political considerations: e.g. Canada carbon tax repeal, Inflation Reduction Act
 - ▶ Informality: e.g. Kenya's charcoal sector
 - ▶ Institutional constraints: e.g. enforcement
 - ▶ Private sector: e.g. carbon offset markets

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 - ▶ Institutional constraints: e.g. enforcement
 - ▶ Private sector: e.g. carbon offset markets
- ▶ **Objective:** maximize the value generated by adoption, given a budget

$$\max \int_{\underline{\alpha}}^{\infty} \mathcal{V}(\alpha) g(\alpha) d\alpha$$

Optimal policy: Three principals

$\tilde{x}^* + \alpha = \text{usage}$

$\pi = \text{fuel savings}$

$\phi = \text{environmental benefit}$

1. Environmental principal:

$$\mathcal{V}^E(\alpha) = (\tilde{x}^* + \alpha)\phi$$

- ▶ Values externalities only: maximize CO₂e abatement
- ▶ 'Inframarginal transfers are wasted subsidy expenditures'
- ▶ Examples: UNFCCC, CORSIA, Article 6.4, carbon credit buyers

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2. Social principal (allocation distortion):

$$\mathcal{V}^A(\alpha) = \mathcal{V}^E(\alpha) + (\tilde{x}^* + \alpha)\pi + \mu(\tilde{x}^*) - p$$

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3. Social principal (resource distortion):

$$\mathcal{V}^R(\alpha) = \mathcal{V}^A(\alpha) - r(p - s_1)$$

- ▶ *Additional cost*: marginal adoptions incur the resource cost
- ▶ *Additional benefit*: inframarginal adopters see a lower resource cost

(1) Distortions increase benefit per subsidy dollar

U = total clean energy usage (proportional to abatement)

$C = s_1 Q + s_2 U$ = subsidy spending

$\tilde{x}^* + \underline{q}$ = usage per marginal adopter

κ = inframarginal cost per marginal adopter

Fixed cost subsidy s_1

$$\frac{dU/ds_1}{dC/ds_1} = \frac{(\tilde{x}^* + \underline{q})}{s_1 + s_2(\tilde{x}^* + \underline{q}) + \kappa}$$

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$$\frac{dU/ds_2}{dC/ds_2} = \frac{\frac{dU}{ds_2}}{U + s_2 \frac{dU}{ds_2} + s_1 g(\underline{q}) \left(-\frac{\partial \underline{q}}{\partial s_2} \right)}$$

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For any given $\{s_1, s_2\}$, both returns rise with $\bar{\gamma}$, through two channels:

1. **Selection:** $\underline{q}$ rises, so marginal adopters abate more
2. **Elasticity:** $|\tilde{\mathcal{E}}(p - s_1)|$ rises $\Rightarrow \kappa$ falls, less inframarginal leakage

(1) Distortions increase benefit per subsidy dollar

$\zeta = Q/\Delta Q =$ subsidy recipients per induced adoption

$\tilde{x}^* + \bar{\alpha} =$ average usage of induced adopters

Welfare gain per fixed cost subsidy dollar, given $\{s_1, s_2\}$, at fixed r :

$$\frac{\Delta W}{\Delta C} = \frac{\mathcal{V}^A(\bar{\alpha}) + r(s_1\zeta - p)}{s_1\zeta + s_2(\tilde{x}^* + \bar{\alpha})}$$

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► **Allocation distortion** increases welfare gain unambiguously:

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 1. **Selection:** $\bar{\alpha}$ rises; induced adoptions generate more welfare
 2. **Elasticity:** ζ falls; less spending to inframarginal adopters
- ▶ **Resource distortion** incurs direct resource cost for induced adopters minus inframarginal adopter resource savings ($rp - rs_1\zeta$)
 - ▶ If $p < s_1\zeta$, resource distortion generates larger welfare gains than allocation distortion
 - ▶ Positive welfare gain as long as selection and elasticity benefits exceed direct cost

(2) Higher $\bar{\gamma}$ tilts the optimal mix toward s_1

Budget constraint: $s_1 \cdot Q + s_2 \cdot U \leq B$

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Equating the s_1 and s_2 FOCs: the optimal split equalizes abatement per dollar
 $\tilde{x}^* + \underline{\alpha}, \tilde{x}^* + \bar{\alpha} = \text{usage by marginal, all adopters}$

$$\begin{array}{c}
 \text{usage per additional adopter} \\
 \overbrace{\tilde{x}^* + \underline{\alpha}} \\
 \hline
 \underbrace{s_1}_{\text{additional cost}} + \frac{p - s_1}{\underbrace{|\tilde{\mathcal{E}}(p - s_1)|}_{\text{inframarginal cost}}} = \frac{\overbrace{d\tilde{x}^*/ds_2}^{\text{additional usage per adopter}}}{\underbrace{(\tilde{x}^* + \bar{\alpha})}_{\text{paid on all usage}} - \underbrace{(1 - \bar{\gamma})(\tilde{x}^* + \underline{\alpha})}_{s_1 \text{ it saves you}}}
 \end{array}$$

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 $\tilde{x}^* + \underline{\alpha}, \tilde{x}^* + \bar{\alpha} = \text{usage by marginal, all adopters}$

$$\underbrace{s_1}_{\text{additional cost}} + \underbrace{\frac{p - s_1}{|\tilde{\mathcal{E}}(p - s_1)|}}_{\text{inframarginal cost}} = \frac{\underbrace{\tilde{x}^* + \underline{\alpha}}_{\text{usage per additional adopter}}}{\underbrace{\frac{d\tilde{x}^*}{ds_2}}_{\text{additional usage per adopter}}} = \frac{\underbrace{(\tilde{x}^* + \bar{\alpha})}_{\text{paid on all usage}} - \underbrace{(1 - \bar{\gamma})(\tilde{x}^* + \underline{\alpha})}_{s_1 \text{ it saves you}}}{1}$$

- ▶ Same impact on elasticity
- ▶ Same impact on selection
- ▶ Marginal cost subsidies are discounted by $\bar{\gamma}$
- ▶ Marginal cost subsidies are higher on higher-usage inframarginal users

Optimal policy: Allowing taxation

► No distortion:

$$s_1^* = 0, \quad s_2^* + \tau^* = \phi$$

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► Allocation distortion: Restores first best

$$s_1^* = p \bar{\gamma} > 0, \quad s_2^* + \tau^* = \phi$$

► Resource distortion: No first best ($h \equiv \frac{g}{1-g}$)

$$s_1^* = \underbrace{\bar{\gamma} \frac{\pi + \phi}{h(\underline{q})}}_{\text{economizes on resource cost}} = r \kappa > 0, \quad s_2^* + \tau^* = \phi$$

Spend the budget reducing the distortion?

In the context of our model:

1. **Correcting an allocation distortion** (e.g. an information campaign):
 $\Rightarrow$ More effective iff its cost per induced adoption $< \kappa$

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 $\Rightarrow$ More effective at equal cost per conversion

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2. **Correcting a resource distortion** (e.g. credit infrastructure):
 $\Rightarrow$ Saves real resources on every adoption, not just at the margin
 $\Rightarrow$ More effective at equal cost per conversion
3. **Subsidizing the resource distortion** (e.g. subsidizing repayment):
 $\Rightarrow$ Always dominated

Today

Introduction

Theory

RCT Design

Impact

Externality of the marginal adopter

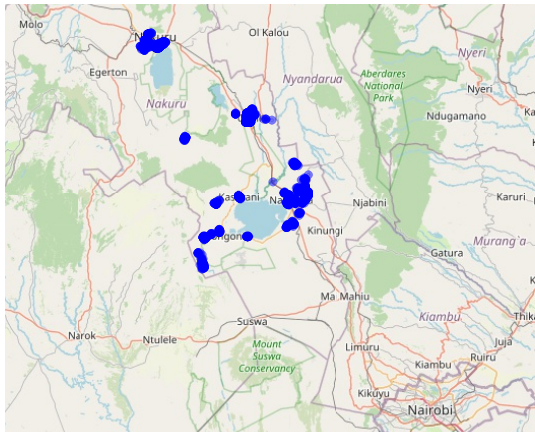
Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

RCT in Nakuru county with 2,134 households

- ▶ Peri-urban regions around Naivasha, Nakuru, Gilgil
- ▶ 65% of households have a grid electricity connection (*Kenya: 58%*)
- ▶ 63% of households cook with charcoal or firewood (*Kenya: 64%*)



2,134 study participants

Enrolled at their homes by Busara enumerators, subject to eligibility criteria:

- ▶ Use traditional charcoal cookstove as primary cooking technology
- ▶ Spend at least \$4 per week on charcoal (5–10% of household income)
- ▶ Have a pre-pay electricity meter

Study participants

Attrition

Randomly assigned credit, FC subsidies, MC subsidies

- ▶ **Fixed cost subsidies**, down from marginal cost of \$82:
 - ▶ 46% assigned a price of \$20 (75% subsidy)
 - ▶ 49% assigned a price of \$73 (10% subsidy)

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 - ▶ 49% assigned a price of \$73 (10% subsidy)
- ▶ **Payment structure**: up front or interest-free four-month loan (\$12 down payment)
- ▶ **Marginal cost subsidies**: 0%, 25%, or 75% subsidy on the cost of electricity used by the induction stove for 6 months after purchase
 - ▶ Implementation similar to Kenya Power subsidy program
- ▶ All treatments fully cross-randomized
- ▶ Treatment assignment is stratified and balanced on observables

Balance

Information about marginal costs of electricity

All flyers

Information (86%)

ECOA YANGU ITATUMIA UNIT/TOKENS NGAPI ZA STIMA KUPIKA?

UGALI



5sh

1sh kwa miezi
6 ya kwanza

SUKUMA WIKI



5sh

1sh kwa miezi
6 ya kwanza

RICE



10sh

3sh kwa miezi
6 ya kwanza

BEANS



25sh

6sh kwa miezi
6 ya kwanza

FISH



5sh

1sh kwa miezi
6 ya kwanza

MAKADIRIO KWA FAMILIA YA WATU 4



ecoa

stoves for life

25sh

6sh kwa miezi
6 ya kwanza

GITHERI



5sh

1sh kwa miezi
6 ya kwanza

TEA



5sh

1sh kwa miezi
6 ya kwanza

MBOGA



5sh

1sh kwa miezi
6 ya kwanza

CHAPATI



20sh

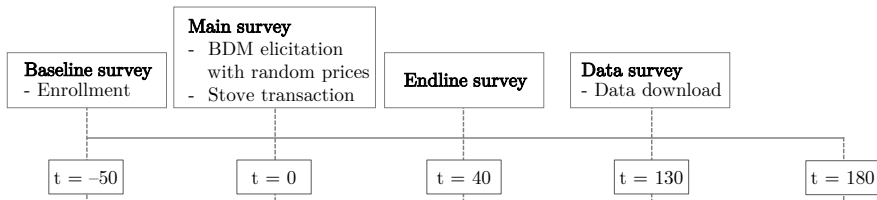
15sh kwa miezi
6 ya kwanza

BEEF STEW



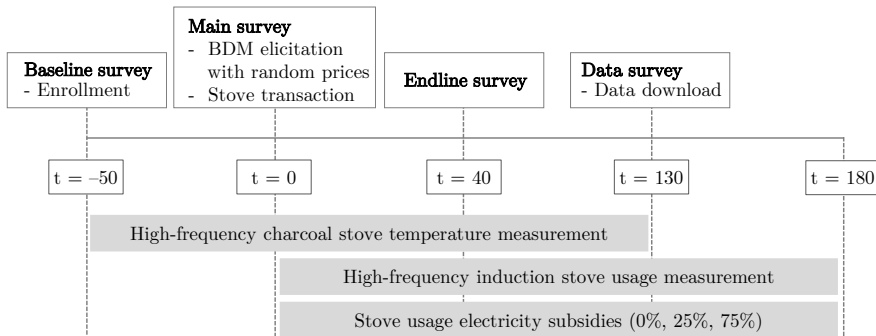
Implementation Timeline

► Implementation: August 2025 – March 2026



Implementation Timeline

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Main visit timeline

1. Collect Kenya Power meter number
2. Inform about electricity subsidy using informational flyer (if relevant) Flyers
3. Inform how credit works (if relevant)

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 - 1) Elicit WTP_i using binary search
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 - 3) WTP is binding: adopt *iff* $WTP_i \geq P_i$
 - ▶ Pay P_i : incentive compatible
 - ▶ 98% of respondents complied

Main visit timeline

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 - Pay P_i : incentive compatible
 - 98% of respondents complied
5. Buyers: demo stove usage, add to customer service WhatsApp group

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Externality of the marginal adopter

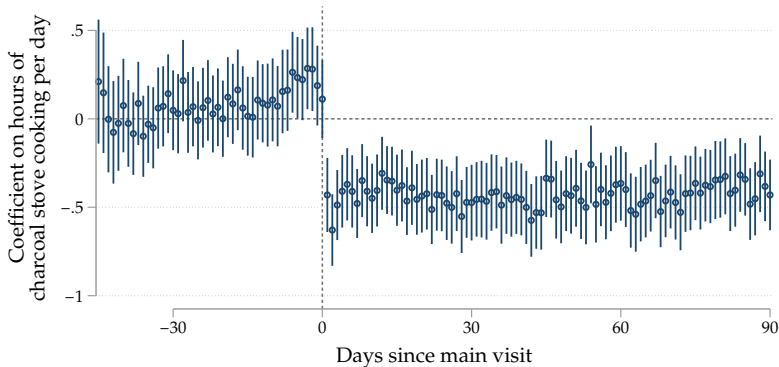
Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

Results: Induction stove adoption reduces charcoal usage

- Data source: temperature sensors
- Control mean: 91 minutes per day
- IV treatment effect: -31 minutes per day



Results: Induction stove adoption reduces energy spending

- Instrumental variables using randomly assigned price as instrument

	Charcoal	LPG	Wood	Electricity	Total
	(1)	(2)	(3)	(4)	(5)
Bought induction stove (=1)	-10.0*** (0.8)	-0.9 (0.6)	-0.3 (0.2)	2.3*** (0.4)	-8.9*** (1.1)
Observations	2081	2081	2081	2081	2081
Control Mean	15.1	5.5	0.7	3.5	24.9
F-Stat	241	241	241	241	241

- Monthly charcoal spending down \$10 (66%)
- Monthly electricity spending up \$2.3 (65%, incl. subsidy)
- Net savings \$8.9 per month
- \$209 over 2.2-year lifespan (10% discount rate), with an \$82 price: 118% ROI

Lifespan

Result: Induction stove adoption reduces CO₂e emissions

- ▶ Instrumental variables using randomly assigned price as instrument
- ▶ Conversion from spending to CO₂ using established methods CO₂ conversion

	Charcoal	LPG	Electricity	Total
Bought induction stove (=1)	-2.50*** (0.20)	-0.03 (0.02)	0.02*** (0.00)	-2.50*** (0.20)
Observations	2081	2081	2081	2081
Control Mean	3.77	0.17	0.04	3.99
F-Stat	241	241	241	241

- ▶ Average abatement across all study participants over the 2.2-year stove life: 5.5 tCO₂e per stove
- ▶ Modest changes in self-reported health symptoms, foods cooked

Fixed subsidy abatement and welfare cost (no loan)

1. **Subsidy cost per tCO₂e**: Each additional adoption costs \$65 in subsidies, abates 5.7 tCO₂e ⇒ **\$11 per tCO₂e**

Alternative assumptions

Fixed subsidy abatement and welfare cost (no loan)

1. **Subsidy cost per tCO₂e**: Each additional adoption costs \$65 in subsidies, abates 5.7 tCO₂e ⇒ **\$11 per tCO₂e**

Alternative assumptions

2. **Observable welfare gain per \$1 spending**: Each \$100 subsidy spending causes 1.5 adoptions, each generating:

Allocation distortion ($r = 0$): \$228 fuel savings, \$82 stove cost, 5.7 tCO₂e

⇒ Only valuing private benefit: \$2.2

⇒ Assuming SCC \$120: \$12.8

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Resource distortion ($r = \frac{\tilde{\gamma}}{1-\tilde{\gamma}}$): as above, minus \$163 in credit costs

⇒ Only valuing private benefit: \$-0.3

⇒ Assuming SCC \$120: \$10.3

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Preferred estimate of distortion ($r = 0.5$): as above, minus \$8.3 in credit costs

⇒ Only valuing private benefit: \$2.1

⇒ Assuming SCC \$120: \$12.6

Resource distortion ($r = \frac{\tilde{\gamma}}{1-\tilde{\gamma}}$): as above, minus \$163 in credit costs

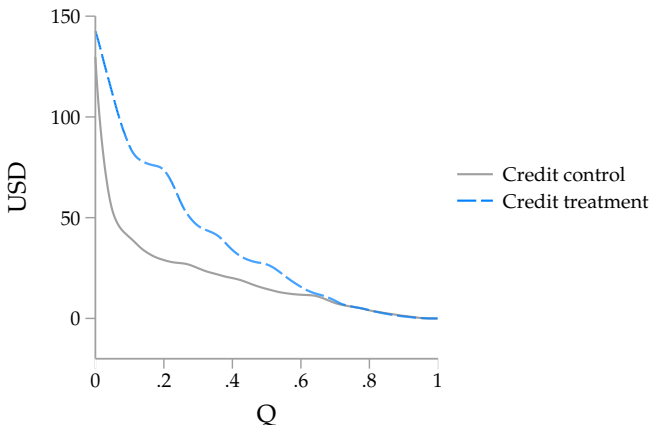
⇒ Only valuing private benefit: \$-0.3

⇒ Assuming SCC \$120: \$10.3

Demand distortions reduce WTP by 45%

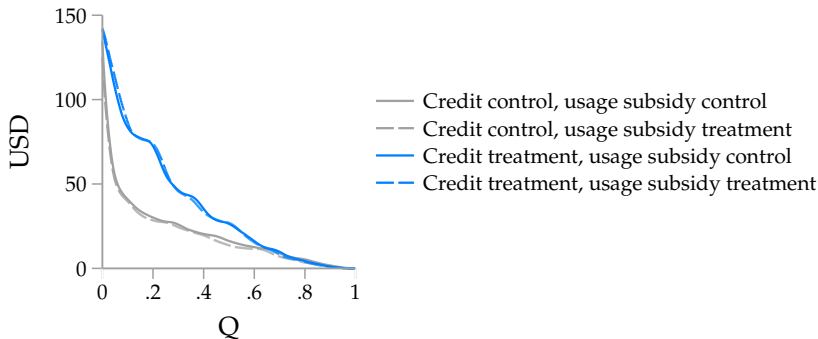
Given \$209 in discounted fuel savings:

- ▶ For participants paying up front, WTP of \$20 implies $\bar{\gamma} = 0.91$
- ▶ For participants paying with loan, WTP of \$35 implies $\bar{\gamma} = 0.83$



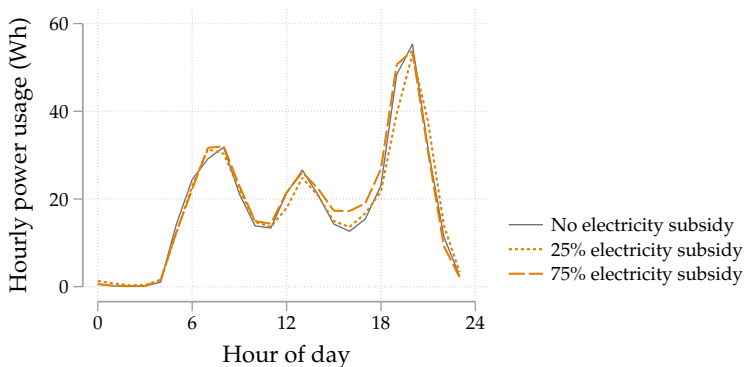
Marginal cost subsidies have no impact on WTP

- ▶ Point estimate: \$0.0 per \$1 subsidy
- ▶ 95% CI: each \$1 of marginal cost subsidy increases WTP by $\leq \$0.2$



Intensive margin is very price inelastic

- ▶ Data source: induction stove 15-minute interval data
- ▶ A 1% price decrease raises usage by $\leq 0.12\%$
- ▶ Randomized SMS reminders have no impact on usage
 $\Rightarrow$ consistent with cooking preferences



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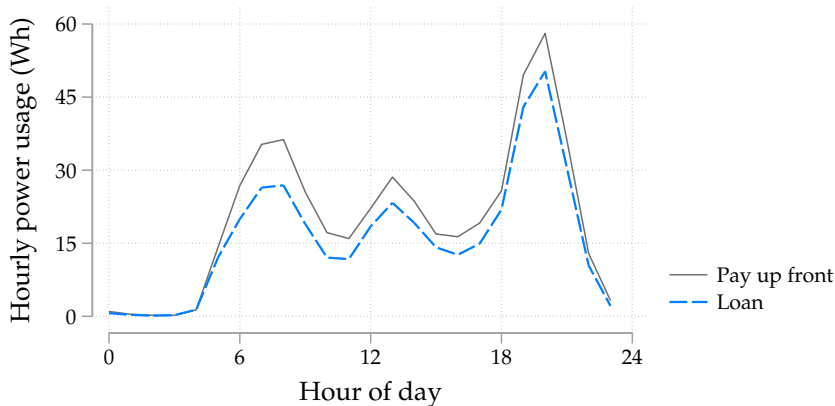
Subsidy expenditure per marginal adoption

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Implications for abatement policy

Marginal adopters with larger distortions use stoves more

- **Theory:** Distortions increase the positive externality generated by marginal adopters
- Define marginal adopters as having $WTP \in [20, 73]$
(IV compliers (marginal adopters): $WTP \in [20, 73]$)



Marginal adopters with larger distortions abate more

	Charcoal			
	(1) USD	(2) USD	(3) CO ₂	(4) CO ₂
Credit (=1)	-1.0 (1.4)	-1.0 (1.4)	-0.2 (0.4)	-0.2 (0.4)
Bought stove (=1)	-10.5*** (0.6)		-2.6*** (0.2)	
Bought X No credit		-10.5*** (0.6)		-2.6*** (0.2)
Bought X Credit	1.6* (0.9)	-8.9*** (0.6)	0.4* (0.2)	-2.2*** (0.2)
Observations	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9

Marginal adopters with larger distortions abate more

	Charcoal				Total			
	(1) USD	(2) USD	(3) CO ₂	(4) CO ₂	(5) USD	(6) USD	(7) CO ₂	(8) CO ₂
Credit (=1)	-1.0 (1.4)	-1.0 (1.4)	-0.2 (0.4)	-0.2 (0.4)	-0.7 (2.0)	-0.7 (2.0)	-0.2 (0.4)	-0.2 (0.4)
Bought stove (=1)	-10.5*** (0.6)		-2.6*** (0.2)		-9.7*** (0.9)		-2.6*** (0.2)	
Bought X No credit		-10.5*** (0.6)		-2.6*** (0.2)		-9.7*** (0.9)		-2.6*** (0.2)
Bought X Credit	1.6* (0.9)	-8.9*** (0.6)	0.4* (0.2)	-2.2*** (0.2)	2.2* (1.2)	-7.4*** (0.8)	0.4* (0.2)	-2.2*** (0.2)
Observations	754	754	754	754	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9	25.5	25.5	4.1	4.1

Summary: Marginal adopters with larger distortions generate larger private and externality benefits

	Loan	Pay up front
	($\tilde{\gamma} = 0.83$)	($\tilde{\gamma} = 0.91$)
Monthly fuel savings among marginal adopters	\$7.4	\$9.7
Advantage of subsidies in a distorted market	+30%	
Annual tCO ₂ e abated per marginal adopter	2.2 tCO ₂ e	2.6 tCO ₂ e
Advantage of subsidies in a distorted market	+19%	

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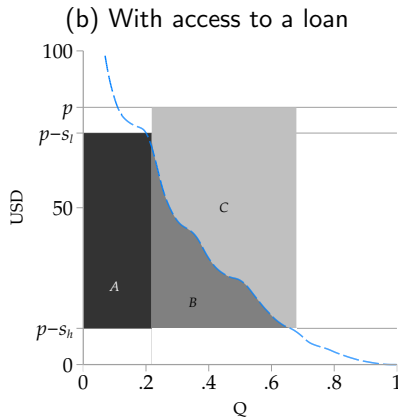
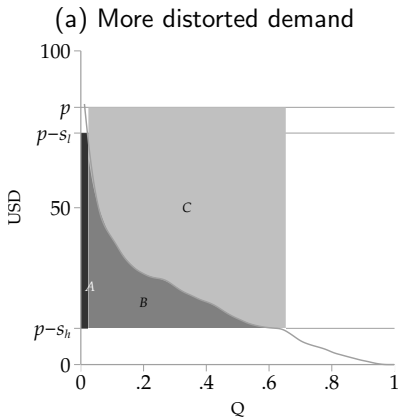
Implications for abatement policy

Demand distortion increases demand elasticity

- ▶ More distorted demand: $\mathcal{E}_D = -2.3$
- ▶ With access to a loan: $\mathcal{E}_D = -0.9$

	Quantity (%)			Log(Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)
Price (USD)	-0.84*** (0.01)	-0.96*** (0.02)	-0.80*** (0.01)			
Log(Price)				-1.23*** (0.01)	-2.33*** (0.02)	-0.88*** (0.01)
Observations	765	387	378	765	387	378
Sample		C=0	C=1		C=0	C=1

More marginal, fewer inframarginal payments



- (A) Inframarginal payments to inframarginal buyers
- (B) Inframarginal payments to marginal buyers
- (C) Marginal payments to marginal buyers

More marginal, fewer inframarginal payments

	Loan ($\bar{\gamma} = 0.83$)	Pay up front ($\bar{\gamma} = 0.91$)
Extensive margin elasticity to fixed subsidy	$\mathcal{E}_d = -0.9$	$\mathcal{E}_d = -2.3$
Fraction of subsidy expenditures inframarginal	38%	6.5%
Subsidy cost per marginal adoption	\$93	\$65
Advantage of subsidies in a distorted market	-30%	

Summary: Effect of demand distortions on subsidy efficacy

	Less distorted ($\bar{\gamma} = 0.83$)	More distorted ($\bar{\gamma} = 0.91$)	Advantage of subsidies in a distorted market
<i>Channel 1: Marginal adopter</i>			
Fuel savings per marginal adoption	\$175	\$228	+30%
tCO ₂ e abated per marginal adoption	4.8 tCO ₂ e	5.7 tCO ₂ e	+19%
<i>Channel 2: Subsidy cost</i>			
Extensive margin elasticity	−0.9	−2.3	
Subsidy spending inframarginal	38%	6.5%	
Subsidy spending per marginal adoption	\$93	\$65	−30%

Summary: Effect of demand distortions on subsidy efficacy

	Less distorted ($\bar{\gamma} = 0.83$)	More distorted ($\bar{\gamma} = 0.91$)	Advantage of subsidies in a distorted market
Subsidy spending per tCO ₂ e abated	\$19	\$11	−41%

Summary: Effect of demand distortions on subsidy efficacy

	Less distorted ($\tilde{\gamma} = 0.83$)	More distorted ($\tilde{\gamma} = 0.91$)	Advantage of subsidies in a distorted market
Subsidy spending per tCO ₂ e abated	\$19	\$11	-41%
Observable welfare gain per \$1 spending			
Allocation distortion ($r = 0$)	\$7.19	\$12.76	+77%
Preferred estimate ($r = 0.5$)	\$7.19	\$12.63	+76%
Resource distortion ($r = \frac{\tilde{\gamma}}{1-\tilde{\gamma}}$)	\$7.19	\$10.27	+43%

Summary: Effect of demand distortions on subsidy efficacy

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What if the distortion was even lower?

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Externality of the marginal adopter

Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

Model estimation: 2-step GMM

- ▶ Private benefit b_i
 - ▶ Distributed Weibull: $b_i \sim \hat{b} \cdot \text{Weibull}(\Omega_1, \Omega_2)$, $\hat{b} = \$209$
- ▶ Social benefit ϕ_i
 - ▶ Function of private benefit: $\phi_i = \hat{\phi} \cdot \left[\Omega_3(b_i/\hat{b}) + (1 - \Omega_3)\varepsilon_i \right]$, $\hat{\phi} = 5.5 \text{ tCO}_2\text{e}$
 - ▶ Mixing parameter $\Omega_3 \in [0, 1]$ governs correlation between b , ϕ

Model estimation: 2-step GMM

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 - ▶ Mixing parameter $\Omega_3 \in [0, 1]$ governs correlation between b , ϕ
- ▶ Heterogeneous demand wedge γ_i
 - ▶ Logit-normal: $\gamma_i = \text{logistic}(\mu_c + \sigma_c \eta_i)$ with $\eta_i \sim N(0, 1)$
 - ▶ Dispersion $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \bar{\gamma}_c)))$
 - ▶ Location μ_c pinned down by $E[\gamma_i|c] = \bar{\gamma}_c$

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We target 7 empirical moments to estimate the 5 parameters

- ▶ Correlation(b , ϕ) (1)
- ▶ Average marginal externality by randomly assigned payment group (2)
- ▶ Extensive margin elasticity by randomly assigned payment group (2)
- ▶ Cost per marginal adoption by randomly assigned payment group (2)

Model fit

Estimated parameters: $\hat{\Omega}_1 = 1.9$, $\hat{\Omega}_2 = 1$, $\hat{\Omega}_3 = 0.6$, $\hat{\Omega}_4 = -2.7$, $\hat{\Omega}_5 = 31$

Moment	Pay up front ($\bar{\gamma} = 0.91$)			Pay with loan ($\bar{\gamma} = 0.83$)		
	Emp.	Model	Dev.	Emp.	Model	Dev.
Correlation (ν)	0.8	0.8	0%	0.8	0.8	0%
Marginal externality (ϕ)	5.7	5.8	1%	4.8	4.9	1%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.4	-1%	-0.9	-0.7	18%
USD per marginal adoption	65.4	64.7	-1%	93.4	97.9	5%

Counterfactual I: Varying demand distortion ($\bar{\gamma}$)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Average demand distortion ($\bar{\gamma}$)	0.95	0.91	0.83	0.75	0.5	0.25	0
Marginal externality (ϕ)	7.8	5.8	4.9	4.5	3.8	3.3	2.8
Extensive margin elasticity ($\mathcal{E}_D$)	-8.9	-2.4	-0.7	-0.4	-0.2	-0.1	-0.1
USD per marginal adoption	62	65	98	143	275	374	458
<i>Observable welfare gain per subsidy dollar (in USD)</i>							
Observed		13	7				
Simulated	20	13	6.9	4.4	1.8	1	0.7
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		11	19				
Simulated	7.9	11	20	32	73	114	162

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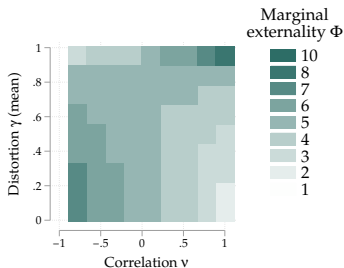
Without demand distortions ($\bar{\gamma} = 0$):

- Welfare gain per subsidy dollar would be **19x lower**
- Abatement cost would be **14x higher**

Counterfactual II: Varying correlation and distortion

(A) Marginal externality

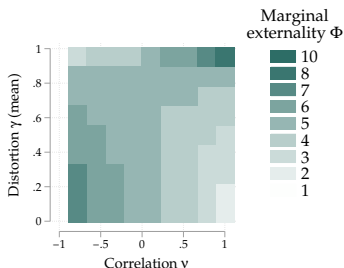
Impact of distortions depends on correlation between WTP and externality



Counterfactual II: Varying correlation and distortion

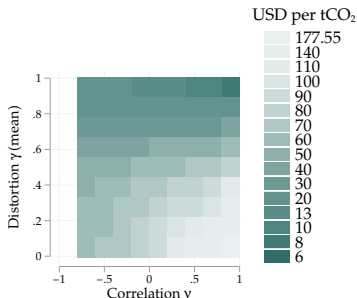
(A) Marginal externality

Impact of distortions depends on correlation between WTP and externality



(B) Abatement cost

Impact of distortions is unambiguous



Result: Demand distortions unambiguously decrease abatement costs in this context, regardless of the correlation coefficient

Selection figure

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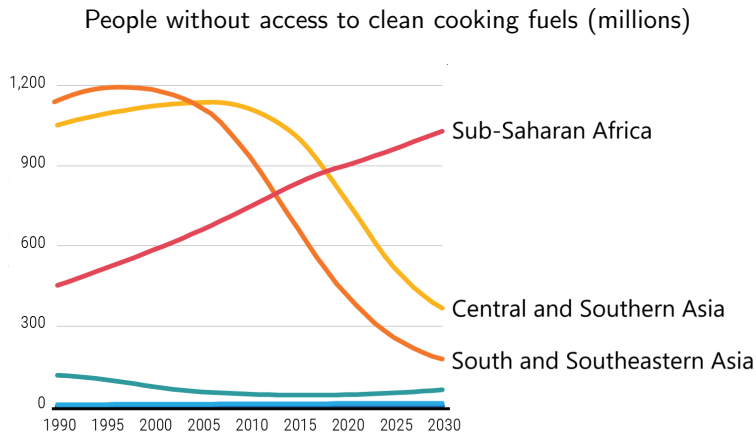
Externality of the marginal adopter

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Implications for abatement policy

2.1 billion people lack clean cooking; >1bn in 2030



Source: Joint SDG 7 Report

Efficiency of selected green subsidies

Technology	Observable welfare gain per US\$1 subsidy (US\$)	Subsidy spend per tCO ₂ e abated (US\$)	Source
<i>(a) Low- and middle-income contexts</i>			
Electric stoves	\$13	\$11	<i>This paper</i>
Solar lanterns	12	10	Rom et al. (2023)
Energy efficient motors	4.1	– ^a	Chaurey et al. (2025)
Improved charcoal stoves	322 ^b	6	Berkouwer & Dean (2026); Hahn et al. (2026)
<i>(b) High-income contexts</i>			
Electric vehicles	0.4	1,356	Hahn et al. (2026)
Electric vehicles	0.3	1,202	Allcott et al. (2026)
Wind production credits	4.9	46	Hahn et al. (2026)
Residential solar	2.9	90	Hahn et al. (2026)
Appliance rebates	0.2	474	Hahn et al. (2026)
Vehicle retirement	0	876	Hahn et al. (2026)
Hybrid vehicles	0	5,940	Hahn et al. (2026)
Weatherization	0	779	Hahn et al. (2026)

Notes: ^aFree dissemination to non-adopters; subsidy spend per abatement not computable.
^bReflects low cost (US\$40), large reductions (2.1 tCO₂e/yr) and savings, and long lifetime.

Conclusion

- **Theory:** Demand distortions increase the efficacy of green subsidies

Randomized trial with 2,134 households in Kenya:

1. Marginal cost subsidies have zero extensive and intensive margin impacts
2. With existing credit constraints, stove subsidies cost \$11 per tCO₂e abated and generate \$12.6 in observable welfare gain per \$1
3. **Testing theory mechanisms:** Demand distortions decrease abatement cost and raise the welfare gain, by:
 - Increasing the marginal adopter's positive externality by 19%
 - Reducing subsidy cost per marginal adopter by 30%
4. **Estimation:** Without demand distortions, abatement cost would be 14x higher and subsidy dollars would generate 19x less welfare
5. **Policy:** Lower portions of the abatement cost curve could be in contexts with larger demand distortions

High LMIC interest rates increase subsidy efficiency

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Source	Year	Country	Loan type	Monthly rate
Karlan, Mullainathan, Roth	2019	India	Moneylenders	141% ^a
Field, Pande	2008	India	Microfinance	44%
Fink, Jack, Masiye	2020	Malawi	Informal, stated	30–40%
Karlan, Mullainathan, Roth	2019	Philippines	Moneylenders	13%
Karlan, Zinman	2008	South Africa	Formal	8–12%
Mckenzie, Woodruff	2008	Mexico	Micro-lenders	10%
Angelucci, Karlan, Zinman ^b	2015	Mexico	Microlending	9%
Egger et al	2022	Kenya	Both	6%
Aleem	1990	Pakistan	Moneylenders	5%
Banerjee, Duflo	2007	India	Both	3–4%
Riley	2024	Uganda	Microfinance	3%
Mckenzie, Woodruff	2004	Sri Lanka	Informal	3%
Banerjee, Duflo	2004	India	Both	2.5–5%
Fafchamps	2000	Kenya, Zimbabwe	Both	2.5–5%
Attanasio et al ^b	2015	Mongolia	Microcredit	2%
Banerjee, Duflo, Glennerster, Kinnan ^b	2015	India	Microfinance	2%
Tarozzi, Desai, Johnson ^b	2015	Ethiopia	Microfinance	1%
Crepon, Devoto, Duflo, Pariente ^b	2015	Morocco	Microfinance	1%

^a4.7% per day; ^bPart of the *AEJ: Applied Economics* special issue on microcredit

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Even higher unrealized **returns** across settings

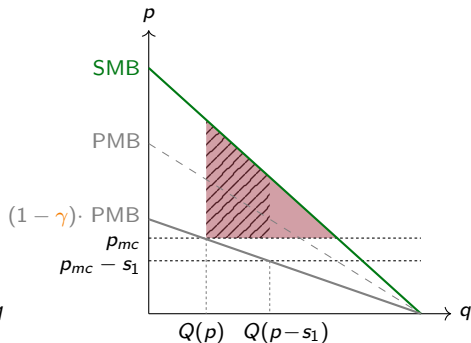
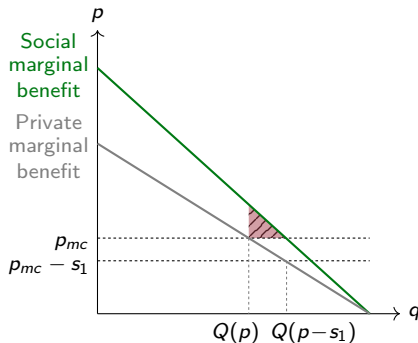
Source	Year	Country	Monthly returns
Banerjee, Duflo	2012	India	89%
Berkouwer, Dean	2022	Kenya	82%
Mckenzie, Woodruff	2008	Mexico	20–33%
Anagol, Udry	2006	Ghana	20–25%
Crepon, Devoto, Duflo, Pariente	2015	Morocco	20%
Field, Pande, Papp, Rigol	2013	India	11%
Kremer, Lee, Robinson, Rostapshova	2013	Kenya	8%
Banerjee, Duflo	2004	India	8%
Dupas, Robinson	2013	Kenya	6%
De Mel, McKenzie, Woodruff	2008	Sri Lanka	5%
Anagol, Udry	2006	Ghana	5%
Fink, Jack, Masiye	2020	Malawi	>4.5%
Banerjee & Duflo	2005	Growth theory: heterogeneous returns to the same factor	

How demand distortions affect deadweight loss

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(a) Without distortion ($\gamma = 0$)

(b) With demand distortion ($\gamma > 0$)



Deadweight loss without subsidy

Deadweight loss removed by a subsidy s_1

Distortions increase the deadweight loss at equilibrium *and* the deadweight loss removed by a fixed cost subsidy.

How correlation generates beneficial vs. adverse selection

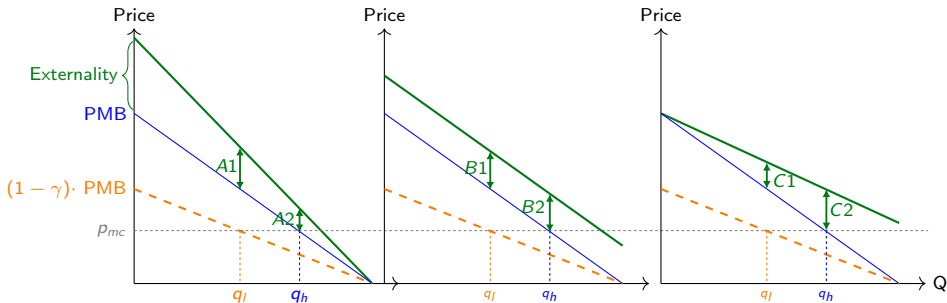
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(a) Positive correlation
($\nu = 1$)

(b) No correlation
($\nu = 0$)

(c) Negative correlation
($\nu = -1$)



- ▶ With no distortion: positive correlation causes *adverse* selection ($A2 < C2$)
- ▶ With distortion: positive correlation causes *beneficial* selection ($A1 > C1$)
- ▶ Distortions increase marginal externality when $\nu > 0$ ($A1 > A2$), decrease it when $\nu < 0$ ($C1 < C2$)

Heatplot

Full model setup

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Building on Allcott, Mullainathan, Taubinsky (2014)

- ▶ Each individual has inelastic energy services demand θ_i (e.g. cooking, driving)
- ▶ At $t = 1$ they decide whether to buy the greener technology, $d_{1i} = \{0, 1\}$
- ▶ At $t = 2$ they choose what fraction $d_{2i} \in [0, 1]$ of θ_i to consume using the greener substitute
- ▶ Usage benefit: $f(d_{2i}, \pi', \theta_i, \alpha_i) = d_{2i}\theta_i\pi' + \mu(d_{2i}\theta_i - \alpha_i)$
 - ▶ $\mu'' < 0$ (diminishing returns to switching); $\mu(0) = 0$
 - ▶ Optimal usage: $\mu'(x^*) = -\pi'$ where $\pi' = \pi + \tau + s_2$
- ▶ Higher taste for clean technology (α_i) means...
 - ▶ ... larger externality reduction: $\phi_i = (x^* + \alpha_i)\bar{\phi}$
 - ▶ ... higher WTP: $wtp_i = (x^* + \alpha_i)\pi' + \mu(x^*) = a + \alpha_i\pi'$
- ▶ Composite demand distortion: $\gamma = \frac{\rho+r}{1+r}$ (allocation ρ , resource r)
- ▶ Market demand: $Q(p - s_1) = \sum_i \mathbb{1}[(1 - \gamma) \cdot wtp_i \geq p - s_1]$

2,134 study participants

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Enrolled at their homes by Busara enumerators, subject to eligibility criteria:

	Mean	25 th	50 th	75 th	N
Number of residents	5.0	4	5	6	2134
Age	35.8	28	35	42	2134
Male (=1)	0.1	0	0	0	2134
Charcoal cooking time (minutes/day)	186.0	140	180	220	2134
Charcoal spending (USD/month)	19.4	14	16	23	2134
LPG spending (USD/month)	9.1	8	8	9	1228
Respondent income (USD/month)	101.5	46	69	116	2127
Household income (USD/month)	246.9	116	196	300	1815
Hours of power outage (30d)	13.9	1	4	12	2134
Hours of voltage fluctuation (30d)	2.0	0	0	1	2134

Balance on baseline characteristics

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	Sample Mean	Credit Treatment	Low BDM Price (i30)	Electricity Treatment	N
	(1)	(2)	(3)	(4)	
Stratification variable: Charcoal spending (USD/month)	19.43 [6.20]	0.23 (0.27)	0.02 (0.27)	-0.04 (0.28)	2134
Stratification variable: Use lpg/gas/ethanol for cooking (=1)	0.58 [0.49]	0.00 (0.02)	0.00 (0.02)	0.00 (0.02)	2134
Stratification variable: Electricity top-up spending (USD, 5 weeks)	4.68 [4.32]	0.39** (0.19)	-0.11 (0.19)	-0.20 (0.20)	2134
Household size	4.95 [1.65]	0.01 (0.07)	0.08 (0.07)	0.03 (0.08)	2134
Age	35.84 [10.15]	0.06 (0.44)	-0.26 (0.44)	-0.18 (0.47)	2134
Male respondent	0.10 [0.30]	0.00 (0.01)	0.02 (0.01)	0.01 (0.01)	2134
Charcoal cooking time (minutes/day)	186.00 [71.08]	-2.02 (3.08)	-3.09 (3.08)	3.00 (3.26)	2134
LPG/ethanol spending (USD/month)	5.23 [10.16]	0.54 (0.44)	-0.49 (0.44)	-0.06 (0.47)	2128
Respondent income (USD/month)	101.44 [115.35]	2.33 (5.00)	1.09 (5.01)	-2.86 (5.29)	2127
Household income (USD/month)	246.99 [204.86]	16.12* (9.61)	16.38* (9.62)	-12.88 (10.18)	1815
Blackout hours (last 30 days)	13.93 [27.49]	2.24* (1.19)	0.50 (1.19)	-0.44 (1.26)	2134
Voltage fluctuation hours (last 30 days)	2.04 [10.88]	-0.70 (0.47)	-0.44 (0.47)	0.20 (0.50)	2134
Joint F-Test p-Value		0.09	0.09	0.21	

Attrition by treatment and adoption

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2,134 respondents completed the main survey; 2,081 completed the endline survey.

Predictors of attrition		
	Control Mean	Attrited (Visit 3)
Credit treatment (=1)	0.030	-0.010 (0.007)
BDM price $\geq$ 30 USD	0.027	-0.005 (0.007)
Electricity treatment $\geq$ 0	0.021	0.006 (0.007)
Bought induction stove (=1)	0.031	-0.022*** (0.007)
Joint F-Test p-Value		0.00
Sample Mean		0.02

- ▶ Attrition balanced across treatments
- ▶ Adopters 2.2pp less likely to attrit; 68% of attriters relocated outside study area

Attrition by treatment and adoption

Back

2,134 respondents completed the main survey; 2,081 completed the endline survey.

Reason for attrition	
Reason	Frequency
Completed survey	2081
Unavailable	8
Withdrew from study	8
Relocated outside survey team reach	36
Hospitalized	1
Total	2134

- ▶ Attrition balanced across treatments
- ▶ Adopters 2.2pp less likely to attrit; 68% of attriters relocated outside study area

Information about marginal costs of electricity

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No information (14%)

NI NINI NAWEZA KUPIKA KUTUMIA ECOA YANGU?



UGALI



SUKUMA WIKI



RICE



BEANS



FISH



ecoq[®]
stoves for life

GITHERI



TEA



MBOGA



CHAPATI



BEEF STEW



Information about marginal costs of electricity

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Information (86%)

ECOQ YANGU ITATUMIA UNIT/TOKENS NGAPI ZA STIMA KUPIKA?

MAKADIRIO KWA FAMILIA YA WATU 4



ecoq
stoves for life



Information about marginal costs of electricity

Back

Information (86%)

ECOA YANGU ITATUMIA UNIT/TOKENS NGAPI ZA STIMA KUPIKA?

UGALI



5sh

1sh kwa miezi
6 ya kwanza

SUKUMA WIKI



5sh

1sh kwa miezi
6 ya kwanza

RICE



10sh

3sh kwa miezi
6 ya kwanza

BEANS



25sh

6sh kwa miezi
6 ya kwanza

FISH



5sh

1sh kwa miezi
6 ya kwanza

MAKADIRIO KWA FAMILIA YA WATU 4



ecoa

stoves for life

25sh

6sh kwa miezi
6 ya kwanza

GITHERI



5sh

1sh kwa miezi
6 ya kwanza

TEA



5sh

1sh kwa miezi
6 ya kwanza

MBOGA



5sh

1sh kwa miezi
6 ya kwanza

CHAPATI



20sh

15sh kwa miezi
6 ya kwanza

BEEF STEW



Information about marginal costs of charcoal

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No information (14%)

WHAT CAN I USE MY CHARCOAL JIKO TO COOK?



UGALI



SUKUMA WIKI



RICE



BEANS



FISH



GITHERI



TEA



MBOGA



CHAPATI



BEEF STEW



Information about marginal costs of charcoal

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Information (86%)

HOW MANY TINS OF CHARCOAL DOES MY JIKO USE TO COOK?

ESTIMATES FOR A HOUSEHOLD OF 4 PEOPLE

	UGALI		15 KSH
	SUKUMA WIKI		10 KSH
	RICE		30 KSH
	BEANS		45 KSH
	FISH		10 KSH



70 KSH		GITHERI	
15 KSH		TEA	
10 KSH		MBOGA	
20 KSH		CHAPATI	
25 KSH		BEEF STEW	

Converting energy usage to CO₂e emissions

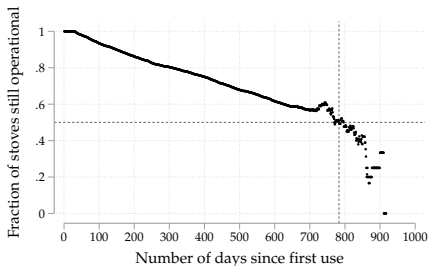
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- ▶ **Charcoal:** (MoFuSS; Bailis et al.; Gill-Wiehl et al.; Floess et al.)
 - ▶ Calorific value: 0.027 TJ per tonne of charcoal
 - ▶ Emissions factor: 161 tCO₂e/TJ combining CO₂, CH₄, N₂O
 - ▶ Fraction non-renewable biomass (fNRB) 38%
 - ▶ 4.3 kgCO₂e per kg charcoal (mean: 3.1 kg charcoal per day, 4.9 tCO₂e per year)
 - ▶ Repeated market price survey ⇒ \$0.21 per kg charcoal
- ▶ **Electricity:** 85% of Kenya's grid is renewable (Kenya Power, 2023)
 - ▶ Average grid emissions intensity: 154 gCO₂e per kWh
 - ▶ Robustness with U.S. grid intensity: 366 gCO₂e per kWh
- ▶ **LPG:** Standard conversion from expenditures to cylinders to CO₂e
- ▶ **Firewood:** 17% of households report using firewood; not tracked
⇒ abatement estimates are conservative

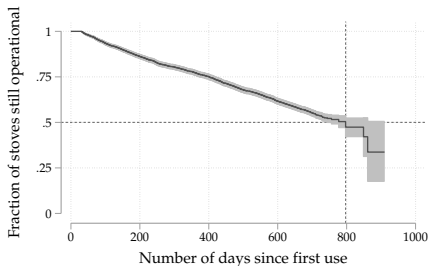
Stove lifespan: 3,797 stoves, Cox model

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(a) Empirical survival



(b) Estimated survival function



- ▶ 3,797 stoves sold on or before 31 Dec 2024 (real-time usage data)
- ▶ Time variable: days between first and last observed usage
- ▶ Breakage: last usage >3 months before end of dataset
- ▶ Cox proportional hazard model $\Rightarrow$ median lifespan 2.2 years

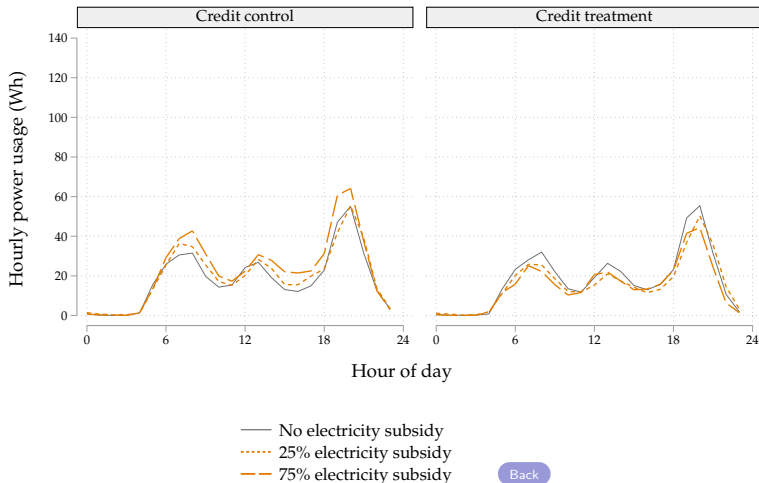
Subsidy cost per tCO₂e under alternative assumptions [Back](#)

Assumption	Cost per tCO ₂ e
Benchmark	US\$11
(1) One-year durability	US\$25
(2) Three-year durability	US\$8
(3) Fraction non-renewable biomass (fNRB) of 10%	US\$14
(4) Fraction non-renewable biomass (fNRB) of 100%	US\$8
(5) US grid emissions intensity	US\$12

- ▶ fNRB matters less because most charcoal emissions are non-combustion
- ▶ Grid intensity matters little: stove adds ~0.05 tCO₂e at U.S. grid vs. 2.5 tCO₂e saved

Intensive margin is very price inelastic (by credit group)

- Demand elasticity of stove usage to electricity price: -0.03



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